

SwasthyaSignal AI

**Actionable Patient Safety Intelligence from Social
Data**

AI For Bharath Hackathon 2026

1. Introduction

SwasthyaSignal AI is a healthcare decision intelligence platform designed to identify actionable patient safety signals from noisy social data sources such as X (Twitter), Reddit, healthcare forums, and online communities.

The system focuses on transforming unstructured public discussions into validated, prioritized, and explainable healthcare intelligence that supports faster operational decision-making.

Unlike traditional healthcare dashboards that only display data, SwasthyaSignal AI provides:

- signal lifecycle tracking,
- confidence-based prioritization,
- explainable escalation,
- and operational recommendations.

The platform demonstrates how AI-powered operational intelligence can improve healthcare monitoring systems and enable early identification of patient safety risks.

2. Problem Statement

Modern healthcare ecosystems generate massive volumes of patient-generated discussions online. These discussions often contain early indicators of:

- adverse drug reactions,
- treatment dissatisfaction,
- medication side effects,
- and emerging healthcare concerns.

However, existing systems face several limitations:

Existing Challenges

1. High Noise Levels

Large volumes of irrelevant and unstructured social content make signal detection difficult.

2. Lack of Prioritization

Most systems cannot differentiate between low-risk and critical healthcare concerns.

3. No Explainable Escalation

Existing dashboards fail to explain why certain alerts are generated.

4. Delayed Operational Response

Healthcare authorities often receive information too late.

5. Dashboard-Centric Systems

Most solutions focus on visualization instead of actionable intelligence.

As a result, important patient safety signals may remain unnoticed until they become critical.

3. Proposed Solution

SwasthyaSignal AI introduces an AI-driven operational intelligence system capable of:

- filtering noisy healthcare discussions,
- detecting meaningful patient safety signals,
- tracking signal evolution over time,
- prioritizing risks using confidence scoring,
- and generating operational recommendations.

The system converts fragmented social discussions into:

- actionable,
- explainable,
- and operational healthcare intelligence.

The solution is designed as a command-center style monitoring platform for healthcare authorities and decision-makers.

4. Objectives

Primary Objectives

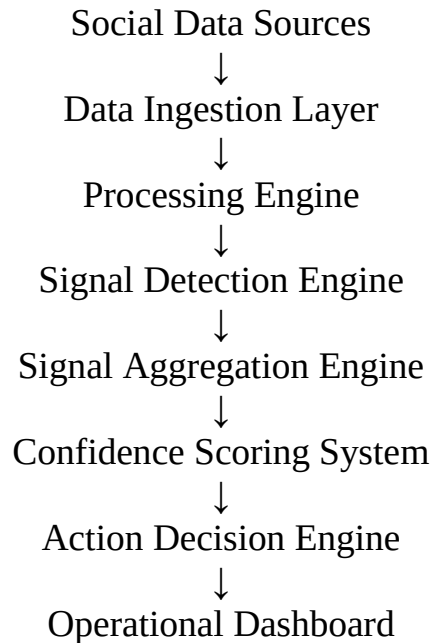
- Detect patient safety risks from social data
- Reduce noise and false alerts
- Enable confidence-based escalation
- Support faster operational decision-making
- Provide explainable healthcare intelligence

Secondary Objectives

- Simulate real-time healthcare monitoring
- Demonstrate scalable signal intelligence architecture
- Improve operational visibility
- Support future healthcare integrations

5. System Architecture

High-Level Architecture



6. Detailed Module Explanation

6.1 Data Ingestion Layer

Description

Simulates continuous healthcare-related social data streams.

Responsibilities

- Stream simulated social posts
- Handle structured/unstructured input
- Feed processing engine

Data Sources

- X (Twitter)
- Reddit
- Healthcare forums
- Community discussions

6.2 Processing Engine

Description

Processes raw social content and extracts meaningful healthcare information.

Functionalities

- Drug extraction
- Symptom identification
- Sentiment analysis
- Adverse event detection

Processing Logic

The engine filters irrelevant information and extracts healthcare-related entities.

6.3 Signal Detection Engine

Description

Detects potential patient safety signals from processed content.

Features

- Signal classification
- Risk categorization
- Frequency analysis
- Severity identification

6.4 Signal Aggregation Engine

Description

Groups related healthcare events together.

Aggregation Factors

- Drug mentions
- Symptom combinations
- Frequency patterns

- Sentiment intensity

Objective

Reduce fragmented alerts and improve operational clarity.

6.5 Confidence Scoring System

Description

Assigns escalation confidence levels based on signal strength.

Signal Lifecycle States

Emerging Signal

Low confidence
→ Monitor

Growing Signal

Medium confidence
→ Investigate

Critical Signal

High confidence
→ Escalate

Importance

Prevents unnecessary escalation while prioritizing critical healthcare risks.

6.6 Action Decision Engine

Description

Generates operational recommendations based on signal confidence.

Decision States

- Monitor
- Investigate
- Escalate

Goal

Provide actionable intelligence instead of raw data visualization.

6.7 Explainability Layer

Description

Displays WHY a signal has been flagged.

Explainability Includes

- confidence score,
- signal frequency,
- severity,
- escalation reason,
- trend evolution.

Importance

Improves trust and transparency for healthcare authorities.

6.8 Dashboard Command Center

Description

Provides centralized operational monitoring.

Features

- Live signal feed
- Alerts panel
- Signal lifecycle tracking
- Monitoring controls
- Interactive dashboard views

7. Functional Requirements

Core Functional Requirements

Signal Monitoring

The system must continuously monitor simulated healthcare discussions.

Signal Detection

The system must identify healthcare-related safety signals.

Signal Aggregation

The system must group related discussions together.

Escalation Logic

The system must classify signals into lifecycle states.

Action Recommendations

The system must generate operational decisions.

Explainability

The system must explain escalation reasoning.

Dashboard Monitoring

The system must display operational intelligence visually.

8. Non-Functional Requirements

Scalability

The architecture should support future real-time integrations.

Reliability

The system should provide deterministic and stable processing.

Performance

Dashboard updates should occur efficiently.

Security

Future implementations should support secure healthcare monitoring.

Maintainability

The modular architecture should support easy upgrades.

Usability

The dashboard should remain simple and operationally focused.

9. Technology Stack

Frontend

- React
- Vite
- JavaScript
- Custom CSS

Intelligence Layer

- Rule-based processing
- Signal aggregation logic
- Confidence threshold engine

Architecture Design

- Component-based UI
- Modular workflow design
- Deterministic simulation logic

10. Workflow Explanation

Complete Workflow

Step 1 — Social Data Stream

The system receives simulated healthcare-related discussions.

Step 2 — Processing

The engine extracts healthcare entities and analyzes sentiment.

Step 3 — Signal Detection

Potential patient safety signals are identified.

Step 4 — Signal Aggregation

Related signals are grouped together.

Step 5 — Confidence Scoring

Escalation confidence is calculated.

Step 6 — Action Generation

The system recommends:

- Monitor
- Investigate
- Escalate

Step 7 — Dashboard Visualization

Operational intelligence is displayed visually.

11. Innovation Highlights

Decision Intelligence

The platform focuses on operational decisions rather than dashboards alone.

Explainable Escalation

Users understand WHY a signal was escalated.

Lifecycle-Based Signal Tracking

Signals evolve dynamically over time.

Command-Center Monitoring

The dashboard is designed as an operational intelligence interface.

Deterministic Simulation

The system provides stable and reliable demo behaviour.

12. Real-World Impact

Healthcare Authorities

Supports early identification of patient safety concerns.

Patients

Enables faster response to adverse events.

Healthcare Systems

Improves operational visibility and reduces noise.

Government Monitoring

Supports scalable healthcare intelligence initiatives.

13. Future Scope

Planned Enhancements

- Real API integrations
- ML-based credibility scoring
- Advanced NLP processing
- Multi-language support
- Healthcare authority integrations
- Real-time monitoring infrastructure

14. Conclusion

SwasthyaSignal AI demonstrates how AI-powered operational intelligence can transform fragmented social healthcare discussions into:

- validated,
- explainable,
- prioritized,
- and actionable healthcare insights.

The platform focuses on operational decision support rather than simple data visualization, making it a scalable and practical healthcare intelligence solution.